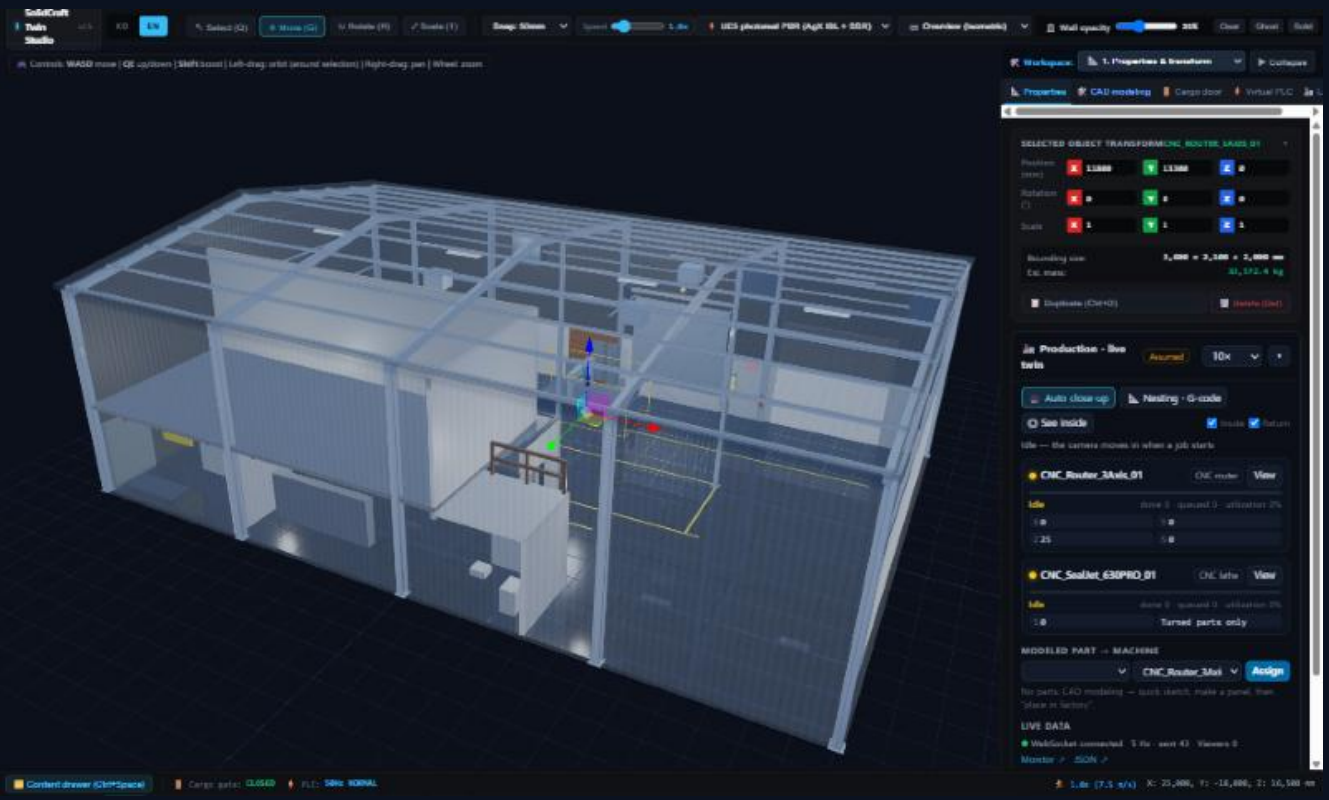
Sugar Hill Factory Digital Twin

User Guide — a friendly walk-through of the 3D factory viewer



Lay out the factory, sketch a part, nest it on a sheet, generate G-code, and watch a virtual CNC router cut it — all in your web browser.

Edition: 9 November 2026 · Applies to the /v2 viewer

Contents

1. Welcome
2. Getting started
3. A tour of the screen
4. Moving around in 3D
5. Selecting, moving and placing objects
6. Editing the building itself
7. Saving and sharing your layout
8. Quick sketch & quote: making a part
9. The live production panel
10. Nesting and G-code
11. Watching the router cut
12. The other sidebar tabs
13. Switching language (KO / EN)
14. Troubleshooting
15. Keyboard shortcuts at a glance
16. What the twin does not do (yet)

1. Welcome

The Sugar Hill Factory Digital Twin is a 3D model of the factory that runs in your web browser. You can walk through the building, rearrange equipment, design simple flat parts, and send them to a virtual CNC router that cuts them exactly as the generated G-code says.

What you can do with it

- **Plan the floor** — place, move, rotate, copy and delete machines and furniture, and even edit walls and partitions.
- **Sketch a part in seconds** — draw an outline, add holes, pick a thickness and get a 3D preview.
- **Nest and generate G-code** — arrange parts on a stock sheet automatically and save a .nc file.
- **Watch production** — see the router load the sheet, clamp it, cut pass by pass, and unload the parts.
- **Share** — save the layout so everyone who opens the same server sees the same factory.

Good to know. Feeds, speeds, cycle times and some machine dimensions are **assumed values** for planning and visualisation. They are marked “Assumed” on screen. Always verify G-code in a real CAM simulator before running it on a machine.

2. Getting started

What you need

- A Windows, macOS or Linux computer with **Node.js** (version 18 or newer).
- A modern browser: Chrome, Edge or Firefox. A screen at least 1600 pixels wide is recommended.
- The project folder (unzip `factory-digital-twin_claude_<date>_vN.zip` — the highest vN is the newest).

Start the twin

1. Open a terminal in the project folder.
2. Install the dependencies (first time only).
3. Start the server.
4. Open **http://localhost:8089** in your browser. This is the hub page; click the **/v2** card (or go straight to **http://localhost:8089/v2**).

```
npm install
npm run dev # serves on port 8089
```

Loading takes a few seconds while the building model is downloaded. When the factory appears and the status bar at the bottom shows *WebSocket connected*, you are ready.

> **Tip.** To stop the server, press **Ctrl+C** in the terminal. To run the automated checks, type `npm test`.

3. A tour of the screen

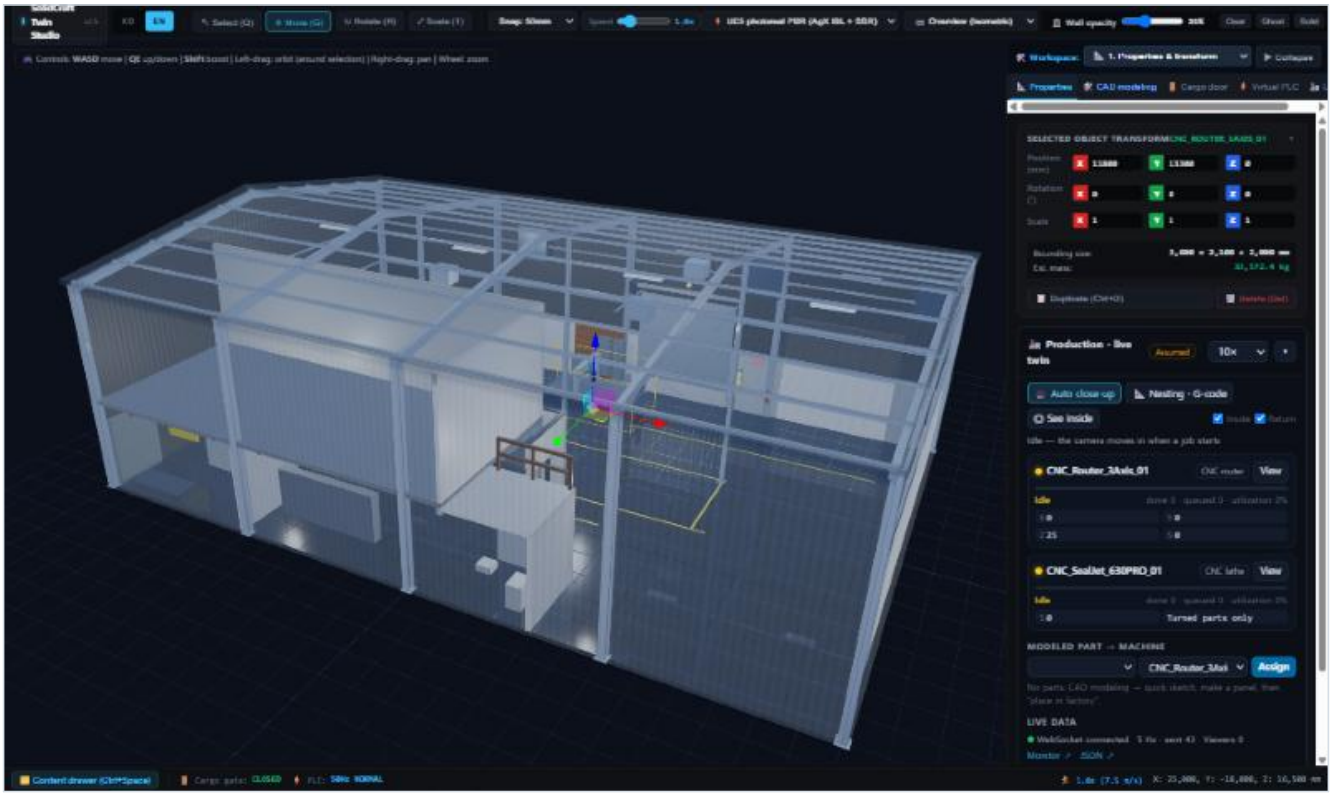


Figure 1 — The /v2 viewer in English mode: toolbar on top, 3D view in the middle, sidebar on the right.

Area	What it is for
Top toolbar (left)	Studio name, the **KO / EN** language switch, and the transform tools: Select, Move, Rotate, Scale, plus snap size and camera speed.
Top toolbar (centre)	Shading mode (photoreal, CAD edges, X-ray, wireframe, splat), camera view presets, and the **Wall opacity** slider with Clear / Ghost / Solid shortcuts.
Top toolbar (right)	Cargo-gate open/close, Undo, **Save**, Cycles render, CAD modeling window, Zen mode and Panel toggle.
3D view	The factory. Click things to select them; drag to look around (see section 4).
Right sidebar	Five tabs: Properties (with the live production panel), CAD modeling, Cargo door, Virtual PLC and Layout.
Bottom bar	Content drawer button (Ctrl+Space), cargo-gate and PLC status, camera speed and the camera position in millimetres.

Good to know. On screens narrower than about 2,000 pixels the right end of the toolbar can be cut off. Use **F4** to hide the sidebar, **Tab** for Zen mode, or widen the browser window.

4. Moving around in 3D

The camera works like a game editor. Hold a key to fly, drag with the mouse to look.

Action	How
Fly forward / back / sideways	`W` `A` `S` `D`
Fly down / up	`Q` / `E`
Fly faster	Hold `Shift`; change base speed with `[` and `]` or the Speed slider
Orbit (look around)	**Left-drag**. If something is selected, the camera orbits around **that object**.
Pan	**Right-drag**
Zoom	**Mouse wheel**
Jump to the selection	`F` (focus)
Jump to an area	Pick a view in the camera dropdown, e.g. *CNC & 3D printer cell*, *1F engineering office*.

Tip. Select a machine first, then left-drag: the view swings around that machine instead of the room centre. It is the quickest way to inspect equipment from every side.

Seeing inside the building

Drag the **Wall opacity** slider down, or click **Clear** (15 %), **Ghost** (40 %) or **Solid** (100 %). The X-ray shading mode shows the interior even more clearly.

5. Selecting, moving and placing objects

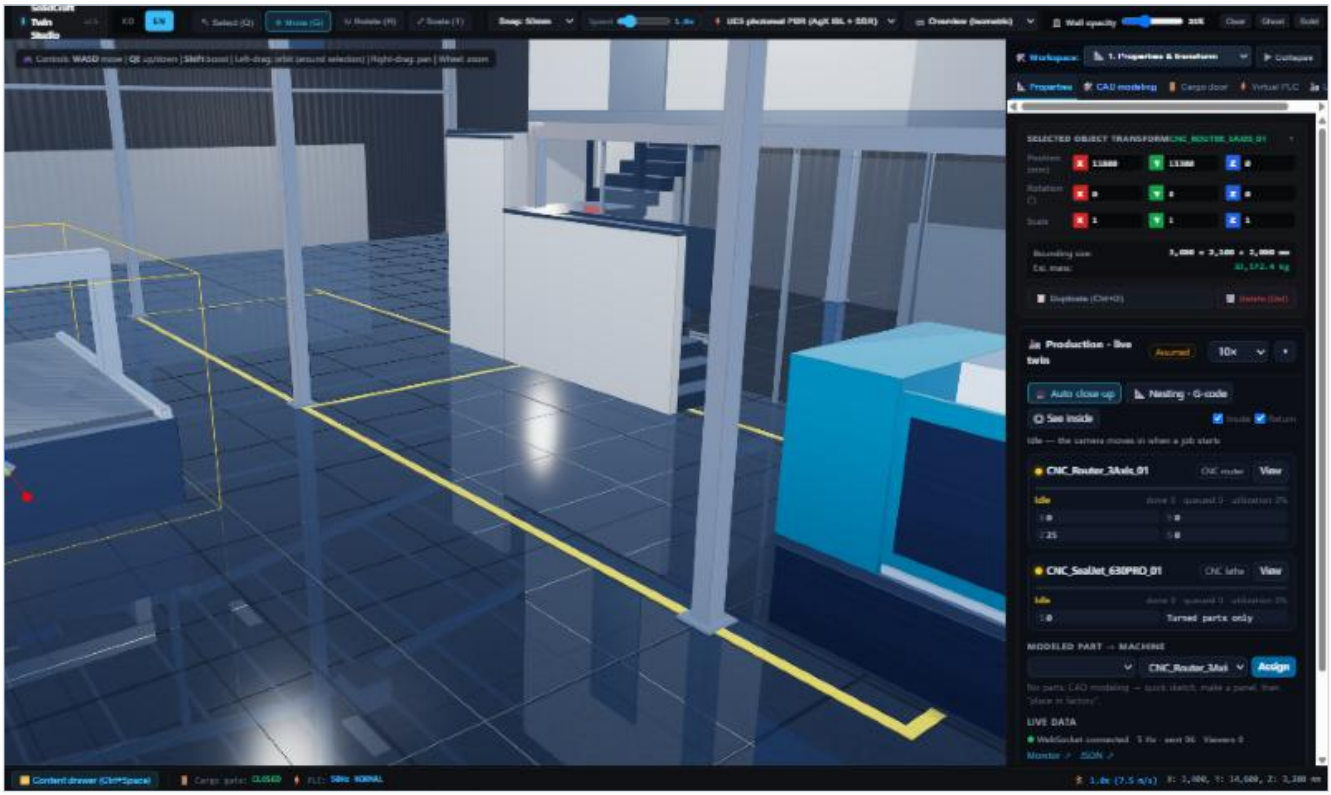


Figure 2 — The CNC router selected (yellow box). Its position, rotation and scale appear in the Properties tab.

Select and transform

- **Click** an object to select it. Its name, position, rotation, scale, size and estimated mass appear in the **Properties** tab. `Esc` clears the selection.
- Choose a tool: Move `G` or `1`, Rotate `R` or `2`, Scale `T` or `3`. Drag the coloured handles (gizmo) in the 3D view.
- For exact values, type numbers into the X / Y / Z boxes in the sidebar (millimetres and degrees).
- The **Snap** dropdown rounds moves to 10, 50, 100 or 500 mm, or turn it OFF for free movement.
- `End` drops the selection onto the floor.

Copy, delete, undo

Action	Key	Button
Duplicate	`Shift+D`	Duplicate (in the Properties tab)
Delete	`Del` or `Backspace`	Delete (in the Properties tab)
Undo	`Ctrl+Z`	Undo (toolbar)

Adding equipment from the content drawer

1. Press `Ctrl+Space` or click **Content drawer** in the bottom-left corner.
2. Pick a category (CNC & 3D printers, Logistics robots, Storage, Office, Safety).
3. Click a card, then click a spot on the floor or a workbench. Press **Esc** to cancel.

> **Good to know.** The default factory starts with only the two CNC machines (router and lathe). Printers, workbenches, tanks and office furniture can be added back from the content drawer at any time.

6. Editing the building itself

Walls, columns, beams, partitions, doors and windows of the building (208 parts) can be edited just like equipment. The floor slab and its joints are locked so you cannot delete the floor by accident.

- **Click** a building part to select it. If a see-through outer wall is in the way, the click goes to the solid part *behind* it.
- **Alt+click** selects the front-most surface, e.g. the transparent outer wall itself.
- Move it with the gizmo, copy it with `Shift+D`, remove it with `Del`, undo with `Ctrl+Z`.
- The Properties tab shows “Building part” as the type, so you always know what you have selected.

> **Tip.** Building edits are kept with the layout. Press **Save** afterwards, otherwise they disappear when the page is reloaded.

7. Saving and sharing your layout

Click **Save** in the toolbar (or *Sync scene layout to server* in the Layout tab). This writes the file `scene_layout.json` in the project folder on the computer that runs the server.

Who sees your change?	Answer
You, after reloading the page	Yes.
Anyone else opening the same server (same computer / network address)	Yes — everyone gets the saved layout.
Someone running their own copy of the project	Only if you give them the updated `scene_layout.json` (it travels inside the project zip or the git repository).

- If `scene_layout.json` has no saved equipment, the built-in default factory is shown.
- **Export JSON** (Layout tab) downloads a copy of the layout for safe keeping.
- **Reset all placed assets** (Layout tab) clears the equipment — use with care, then Save.

8. Quick sketch & quote: making a part

Click the **CAD modeling** tab in the sidebar (or the **CAD modeling** button in the toolbar). The **Quick sketch & quote** window opens straight away.

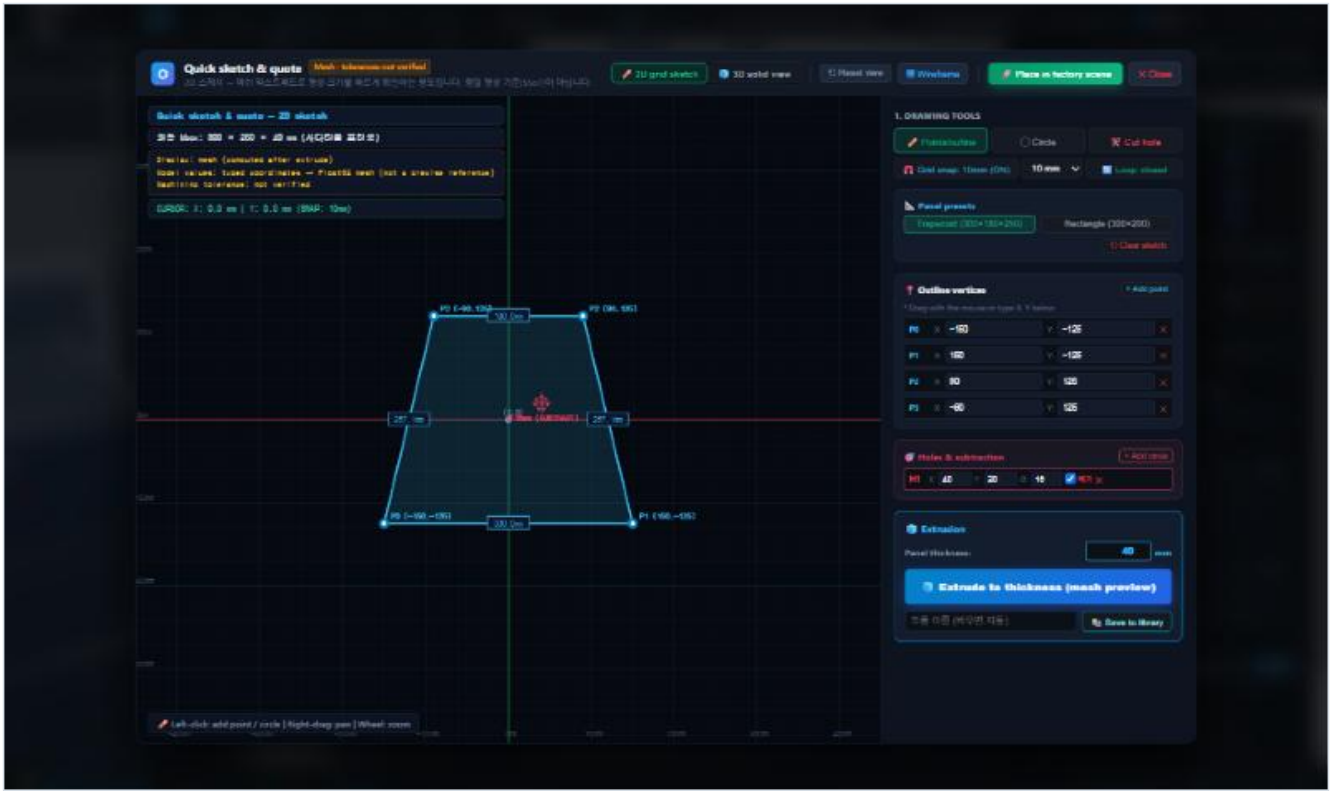


Figure 3 — 2D grid sketch: the trapezoid preset with one hole. Vertices and holes can also be typed in on the right.

Draw the outline

1. Choose a tool: **Points/outline**, **Circle** or **Cut hole**.
2. Left-click on the grid to add corner points (snap is 10 mm by default). Right-drag pans, the wheel zooms.
3. Or start from a preset: **Trapezoid** (300 × 180 × 250) or **Rectangle** (300 × 200).
4. Fine-tune any corner in the **Outline vertices** list, and any hole in **Holes & subtraction** (X, Y, diameter).

Turn it into a 3D part

1. Enter the **Panel thickness** in millimetres.
2. Click **Extrude to thickness** to see the 3D preview.
3. Optionally type a name and click **Save to library** so the part is available for nesting later.
4. Click **Place in factory scene** to put the part into the 3D factory.

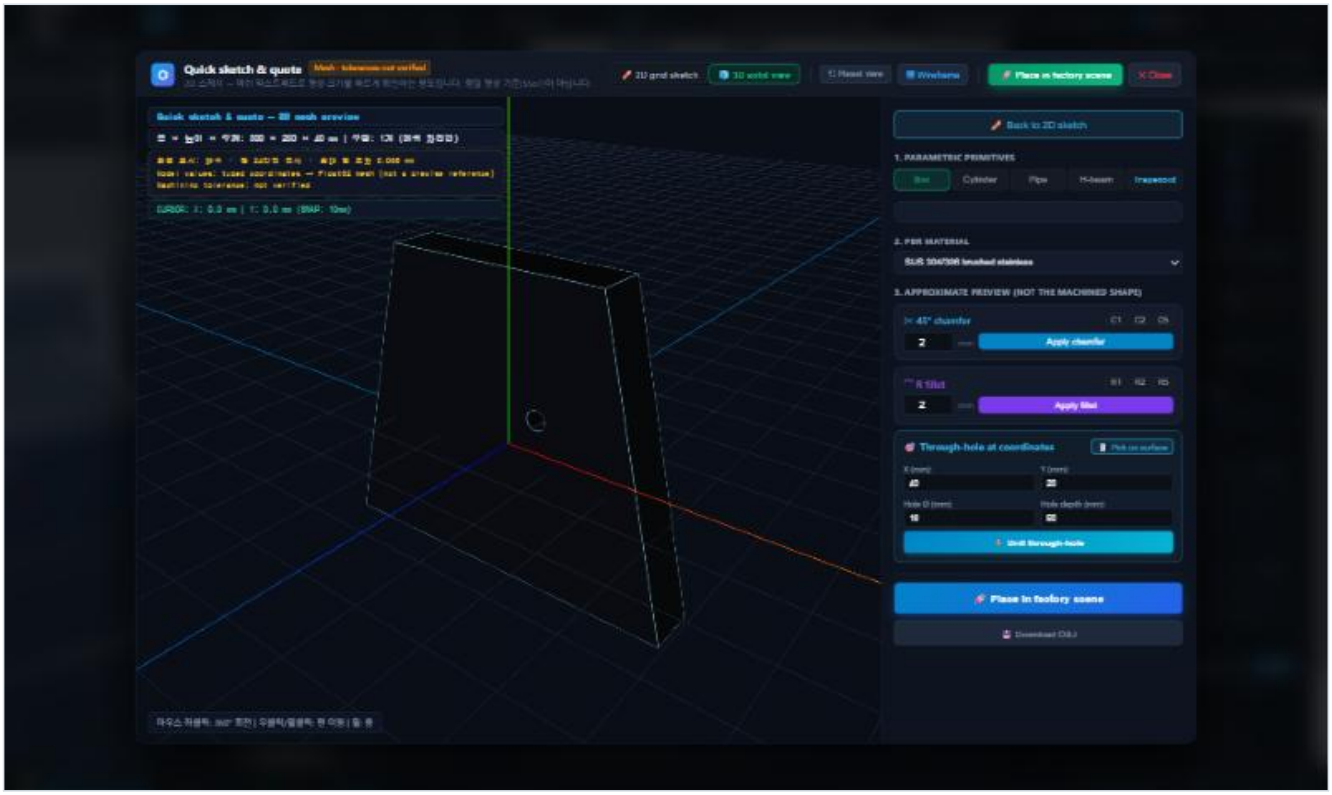


Figure 4 — 3D preview after extrusion, with material choice and approximate chamfer, fillet and drilling tools.

> **Good to know.** The sketch window makes a display mesh, not a precision CAD solid. Chamfer and fillet are approximate previews, and machining tolerances are not verified.

9. The live production panel

In the **Properties** tab, under the selected object, you will find **Production · live twin**. It lists every machine in the factory with its state and axis values.

Control	What it does
Speed (1× … 3600×)	Simulation speed. 10× is a good default; 600× runs a long job in seconds.
Auto close-up	When a job starts, the camera flies to the tool and follows it. Touching the camera pauses the follow; it resumes after 20 s.
Inside / Return	**Inside**: make enclosed machine housings see-through during a close-up. **Return**: go back to your previous view when the job ends.
See inside	Make machine housings see-through right now.
Machine card → View	Jump the camera to that machine (and follow it, if it is busy and auto close-up is on).
Modeled part → machine	Pick a part you placed in the factory and a machine, then click **Assign** to start a job.
Live data	Shows whether the WebSocket telemetry stream is connected and how many monitor screens are watching.

> **Tip.** Click the small arrow in the panel header to fold the panel when you need more room in the sidebar.

10. Nesting and G-code

Click **Nesting · G-code** in the production panel. Tick the parts you want, set quantities, and the layout and G-code update automatically.

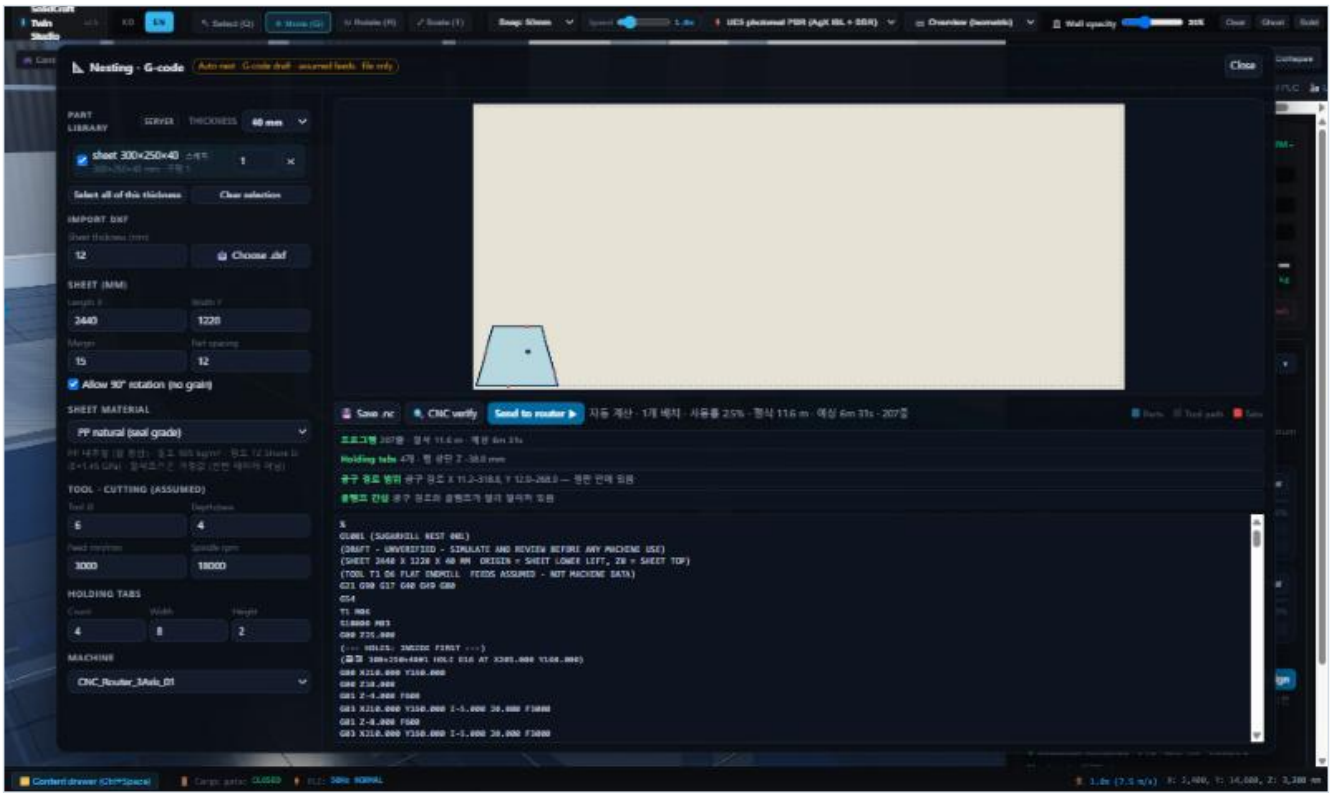


Figure 5 — The nesting window: part library on the left, sheet layout and verification in the middle, G-code below.

Section	Settings
Part library	Your saved parts. Filter by thickness, tick a part and set the count. **Import DXF** adds parts from a drawing.
Sheet (mm)	Stock size (default 2440 × 1220), margin, part spacing, and whether parts may rotate 90°.
Sheet material	Sets the density (for mass) and the suggested cutting values.
Tool · cutting	Tool diameter, depth per pass, feed and spindle speed (assumed values — check them).
Holding tabs	Small bridges that keep parts attached until the last moment: count, width and height.
Machine	Which router receives the job.

Below the sheet you will see the **verification** lines: stays-on-sheet check, holding tabs, tool path bounds and clamp clearance. A red line blocks sending.

- **Save .nc** — download the G-code file (it is a draft: simulate it before real use).
- **CNC verify** — explains how to check the file in the free CAMotics simulator, and can open it directly if CAMotics is configured on the server.
- **Send to router** — starts the job on the virtual router.

11. Watching the router cut

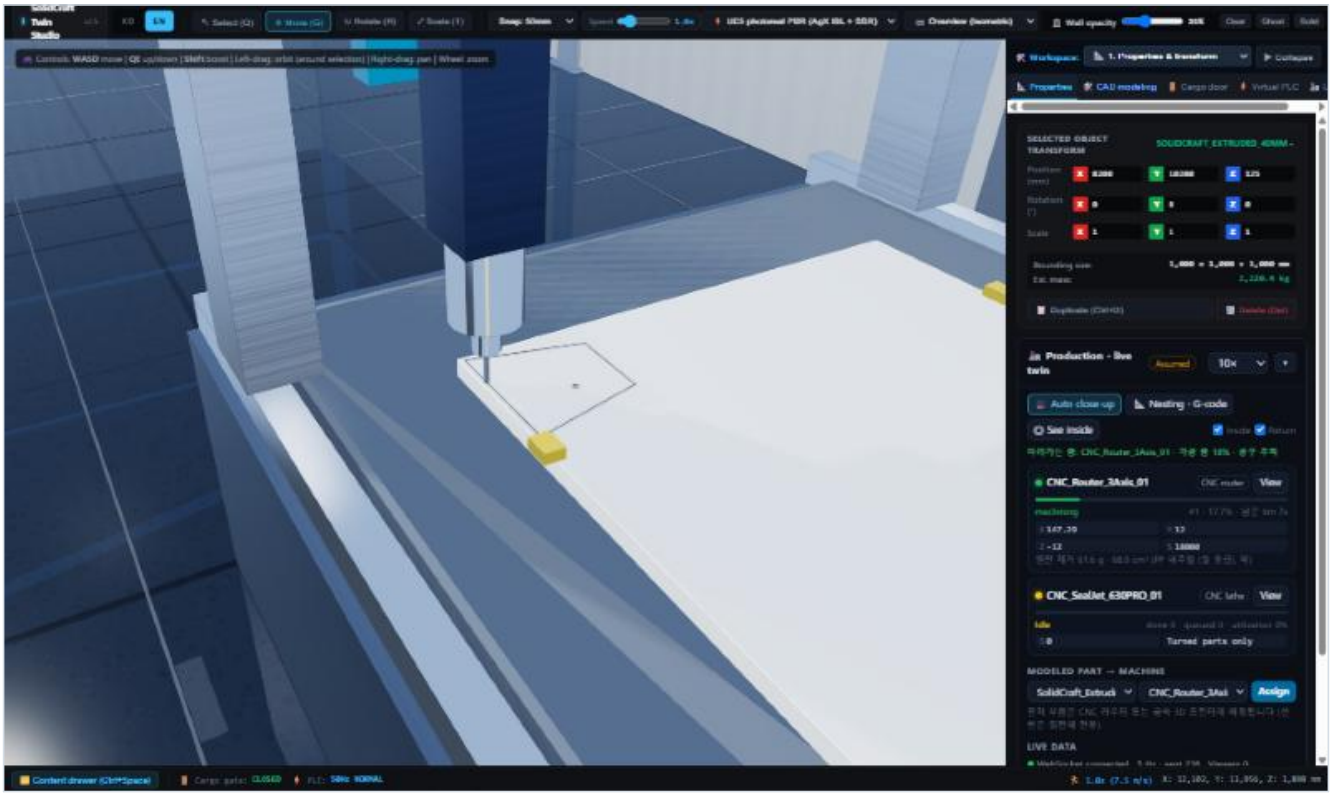


Figure 6 — Auto close-up following the router while it cuts the outline. The yellow blocks are the sheet clamps.

1. **Loading** — the sheet is lowered onto the table, four clamps close and the vacuum switches on.
2. **Running** — the tool replays the G-code line by line. The groove deepens with each pass and darkens with depth; through-cuts turn almost black.
3. **Holding tabs** stay uncut so parts do not move.
4. **Unloading** — clamps open, the finished parts lift out and are placed in front of the machine.

The machine card shows progress, axis positions, and the material removed so far (grams and cm³).

> **Good to know.** Holes are cut first, then the outline, 4 mm deeper on every pass by default. That order mirrors a real router program and is why the holes go all the way through early in the job.

12. The other sidebar tabs

Tab	What you find there
Cargo door	Open / Stop / Close the overhead sectional door and choose which wall it is installed on.
Virtual PLC	Simulated digital inputs and outputs, spindle speed, chamber oxygen and live ladder logic.
Layout	List of placed assets, sync to server, export JSON, and reset.

13. Switching language (KO / EN)

The large **KO | EN** button next to the studio name switches the whole interface between Korean and English. The highlighted side is the current language, and your choice is remembered next time.

> **Good to know.** Main controls, panels and windows are translated. A few long equipment descriptions and log messages stay in Korean rather than being shown half-translated.

14. Troubleshooting

Problem	What to try
The page does not load	Check the terminal: the server must be running (`npm run dev`). Use the address it prints.
My changes are gone after reload	Press **Save** before reloading. Edits are not saved automatically.
I cannot see the Save button	The toolbar may be cut off on a narrow screen. Press **F4** to hide the sidebar or widen the window.
A part will not go to the router	Open the nesting window and read the red verification line — usually the part is larger than the sheet.
Rebuild / Render / CAMotics buttons say “not configured”	These run programs on the server computer and only work when it is set up: SOLIDCRAFT_CLI, BLENDER_PATH or CAMOTICS_PATH must point to the program, and the server must be opened on the same computer (localhost).
The camera keeps moving to a machine	That is **Auto close-up**. Turn it off in the production panel, or simply drag the view to pause it.
I clicked the wall but wanted the machine behind it	Plain click already prefers solid things behind glass. Use **Alt+click** only when you want the front-most surface.

15. Keyboard shortcuts at a glance

Key	Action	Key	Action
W A S D	Fly	G / 1	Move tool
Q / E	Down / up	R / 2	Rotate tool
Shift	Fly faster	T / 3	Scale tool
[/]	Camera speed − / +	F	Focus selection
Ctrl+Space	Content drawer	End	Drop to floor
Tab	Zen mode	Shift+D	Duplicate
F4	Show / hide sidebar	Del	Delete
Esc	Deselect / cancel	Ctrl+Z	Undo

Mouse: **left-drag** orbit (around the selection), **right-drag** pan, **wheel** zoom, **click** select, **Alt+click** select the front-most surface.

16. What the twin does not do (yet)

- Cutting speeds, cycle times and machine dimensions are planning assumptions, not measured values.
- Nesting packs parts by their bounding rectangles, so irregular shapes do not interlock yet.
- G-code is a draft for a Fanuc-style 3-axis router. Simulate it (for example in CAMotics) before any real use.
- The sketch tool produces display meshes; precise B-Rep solids (STEP) are outside this viewer.
- Automatic tool changes are not supported yet.

> **Tip.** Found something confusing or broken? Note what you clicked and what you expected, take a screenshot, and pass it on with the project version number (the _vN in the zip name).